9.1 Smoke and germination

Science as a human endeavour

Verbal/linguistic

Germination is the process of seeds sprouting and growing. For most seeds, germination involves coming into contact with water. The seeds absorb water, swell up and begin growing. But the seeds of some plant species (such as wattles) germinate better after bushfires. Wattles have seeds with tough waterproof coats. Scientists discovered that if this coating is split, water can enter. Experiments showed that soaking wattle seeds in boiling water for a few minutes could split the seed coat. So, for a long time, ecologists thought it was only heat that caused wattles to germinate after a bushfire.

sprout (v) to begin growing; to put out shoots

swell up (v) to expand; to get bigger

bushfire (n) an uncontrolled fire that burns in bushland

Before 1990, many plant species that the horticultural industry and researchers had been trying to grow would not germinate very well. For these 'problem species', heat treatment did not work, and the researchers had no idea why. For example, in Western Australia, about 20% of the native species were difficult to germinate. This meant that these species could not be:

- grown in plant nurseries and sold to the public
- widely used in repairing natural ecosystems, such as after mining.

Anyone who wanted to use these species had to obtain plants and seeds from the wild. This put pressure on the natural ecosystems.

nursery (n) a place where plants are grown to be sold

mining (n) the industry of digging up the earth to get minerals

In 1990, two South African scientists discovered that smoke, rather than heat, improved the germination of some South African plants. The climate, flora and bushfire history of South Africa is similar to that of Western Australia. So scientists at Kings Park and Botanic Gardens (in Western Australia) decided to try smoke on Western Australian plants. They discovered a dramatic response of many species to smoke. This led to more research around Australia. Scientists have now shown that more than 400 species of plants germinate better after smoke treatment. Because of this research, smoke is now widely used in nursery production, bushland management and mine site restoration in Australia. It is even used by some home gardeners and farmers who grow native plants.

bushland management (n) looking after areas of native bush

mine site restoration (n) planting and landscaping an area after mining is finished

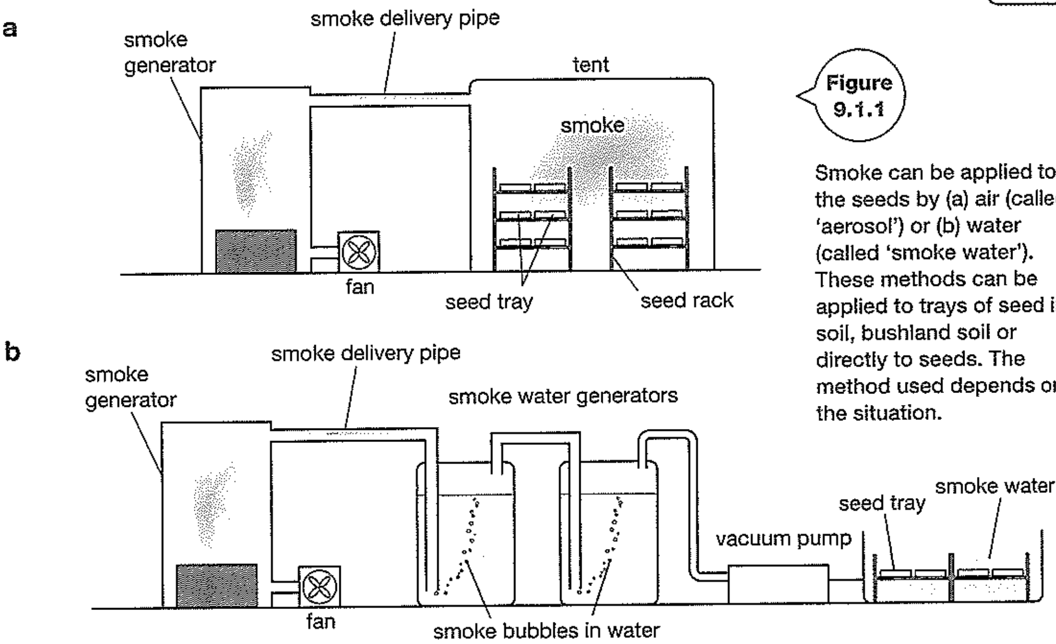

Figure 9.1.1

Smoke can be applied to the seeds by (a) air (called 'aerosol') or (b) water (called 'smoke water'). These methods can be applied to trays of seed in soil, bushland soil or directly to seeds. The method used depends on the situation.

9.1

1 **Define** germination.

2 **Explain** why Australian scientists thought it was probably heat that caused wattles to germinate after a fire.

3 **Explain** why Western Australian scientists did not know how to germinate some 'problem species'.

4 **Describe** three benefits of improving the germination of 'problem species'.

5 **Explain** why scientists at Kings Park and Botanic Gardens decided to try smoke to improve germination of Western Australian native plants.

6 **Describe** the two methods of applying smoke to the seeds.